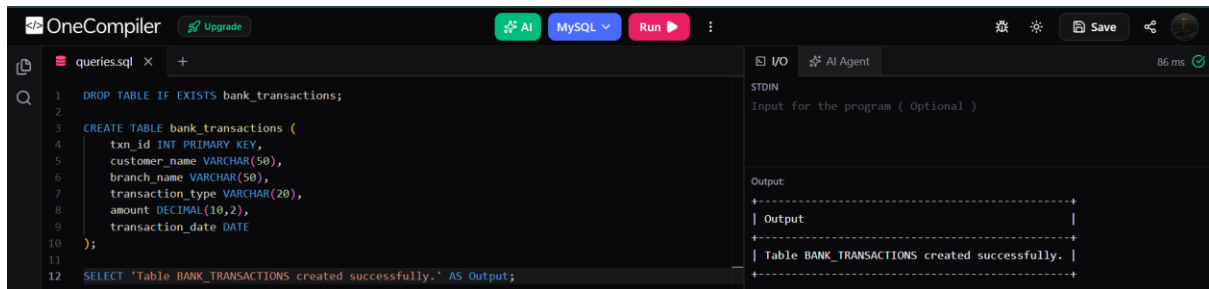


SEC 1: DDL COMMANDS

1. CREATE TABLE — Bank Transactions

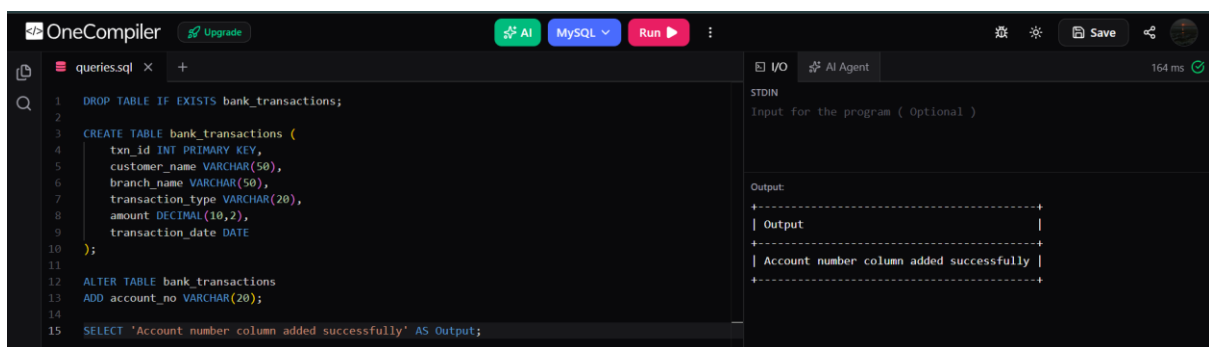


The screenshot shows the OneCompiler interface with a MySQL query editor. The query is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 SELECT 'Table BANK_TRANSACTIONS created successfully.' AS Output;
```

The output on the right shows the message: "Table BANK_TRANSACTIONS created successfully."

2. ALTER TABLE — Add Account Number

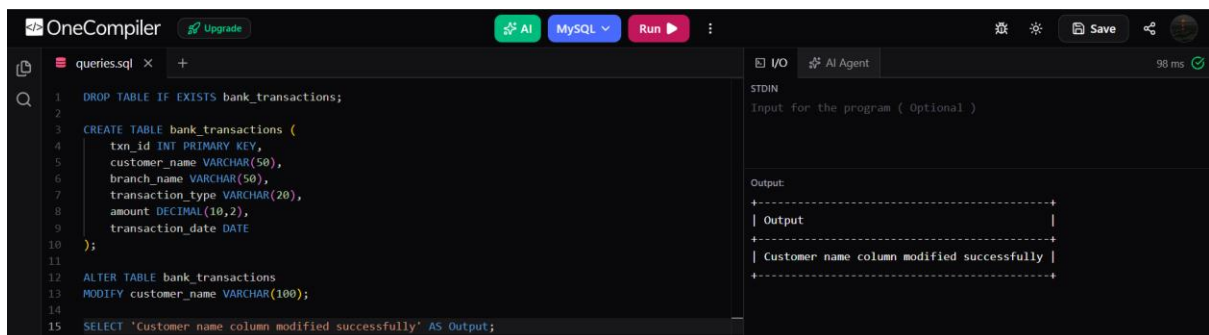


The screenshot shows the OneCompiler interface with a MySQL query editor. The query is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 ALTER TABLE bank_transactions
13 ADD account_no VARCHAR(20);
14
15 SELECT 'Account number column added successfully' AS Output;
```

The output on the right shows the message: "Account number column added successfully"

3. Modify Customer Name Column

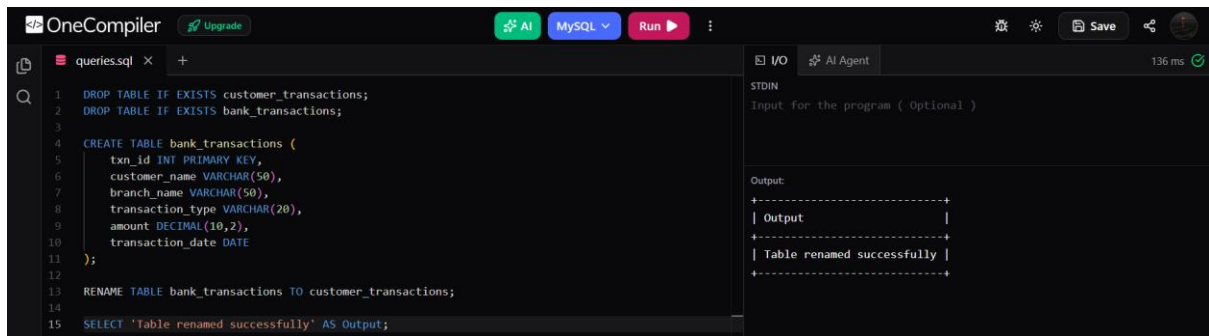


The screenshot shows the OneCompiler interface with a MySQL query editor. The query is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 ALTER TABLE bank_transactions
13 MODIFY customer_name VARCHAR(100);
14
15 SELECT 'Customer name column modified successfully' AS Output;
```

The output on the right shows the message: "Customer name column modified successfully"

4. Rename Bank Transactions Table



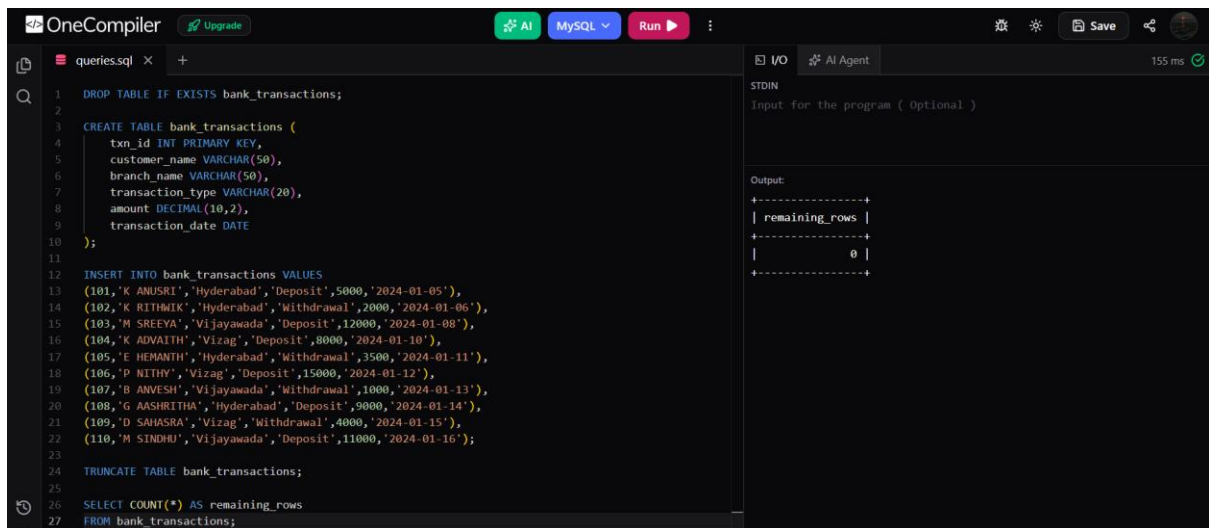
The screenshot shows the OneCompiler IDE with a MySQL query editor. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS customer_transactions;
2 DROP TABLE IF EXISTS bank_transactions;
3
4 CREATE TABLE bank_transactions (
5     txn_id INT PRIMARY KEY,
6     customer_name VARCHAR(50),
7     branch_name VARCHAR(50),
8     transaction_type VARCHAR(20),
9     amount DECIMAL(10,2),
10    transaction_date DATE
11);
12
13 RENAME TABLE bank_transactions TO customer_transactions;
14
15 SELECT 'Table renamed successfully' AS Output;
```

The output panel on the right shows the result of the execution:

```
Output:
+-----+
| Output |
+-----+
| Table renamed successfully |
+-----+
```

5. Truncate Bank Transactions Table



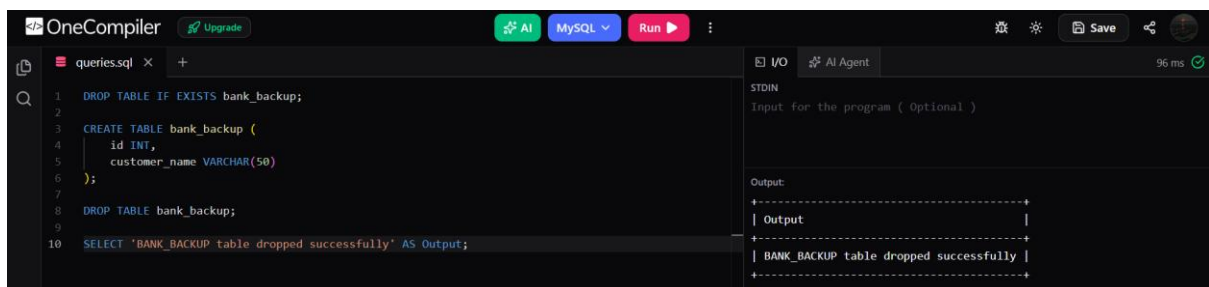
The screenshot shows the OneCompiler IDE with a MySQL query editor. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10);
11
12 INSERT INTO bank_transactions VALUES
13 (101,'K ANUSRI','Hyderabad','Deposit',5000,'2024-01-05'),
14 (102,'K RITHMIK','Hyderabad','Withdrawal',2000,'2024-01-06'),
15 (103,'M SREEYA','Vijayawada','Deposit',12000,'2024-01-08'),
16 (104,'K ADVAITH','Vizag','Deposit',8000,'2024-01-10'),
17 (105,'E HEMANTH','Hyderabad','Withdrawal',3500,'2024-01-11'),
18 (106,'P NITHY','Vizag','Deposit',15000,'2024-01-12'),
19 (107,'B ANVESH','Vijayawada','Withdrawal',1000,'2024-01-13'),
20 (108,'A AASHRITHA','Hyderabad','Deposit',9000,'2024-01-14'),
21 (109,'D SAHASRA','Vizag','Withdrawal',4000,'2024-01-15'),
22 (110,'M SINDHU','Vijayawada','Deposit',11000,'2024-01-16');
23
24 TRUNCATE TABLE bank_transactions;
25
26 SELECT COUNT(*) AS remaining_rows
27 FROM bank_transactions;
```

The output panel on the right shows the result of the execution:

```
Output:
+-----+
| remaining_rows |
+-----+
| 0 |
+-----+
```

6. Drop Bank Backup Table



The screenshot shows the OneCompiler IDE with a MySQL query editor. The code in the editor is as follows:

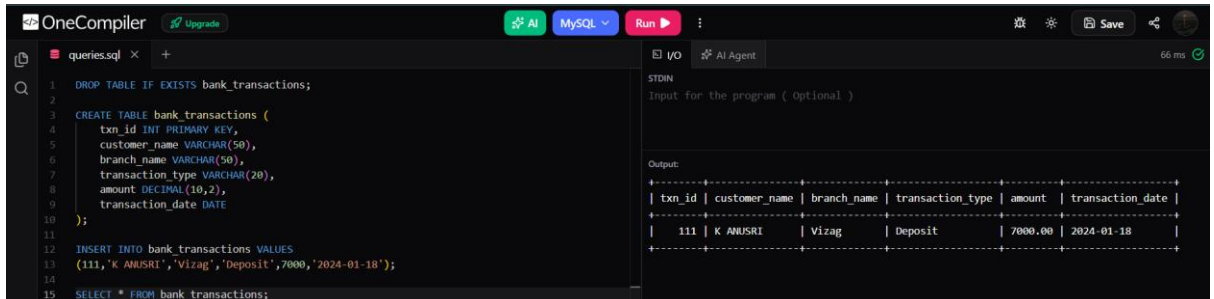
```
1 DROP TABLE IF EXISTS bank_backup;
2
3 CREATE TABLE bank_backup (
4     id INT,
5     customer_name VARCHAR(50)
6 );
7
8 DROP TABLE bank_backup;
9
10 SELECT 'BANK_BACKUP table dropped successfully' AS Output;
```

The output panel on the right shows the result of the execution:

```
Output:
+-----+
| Output |
+-----+
| BANK_BACKUP table dropped successfully |
+-----+
```

SEC 2: DML COMMANDS

1. Insert New Customer Transaction



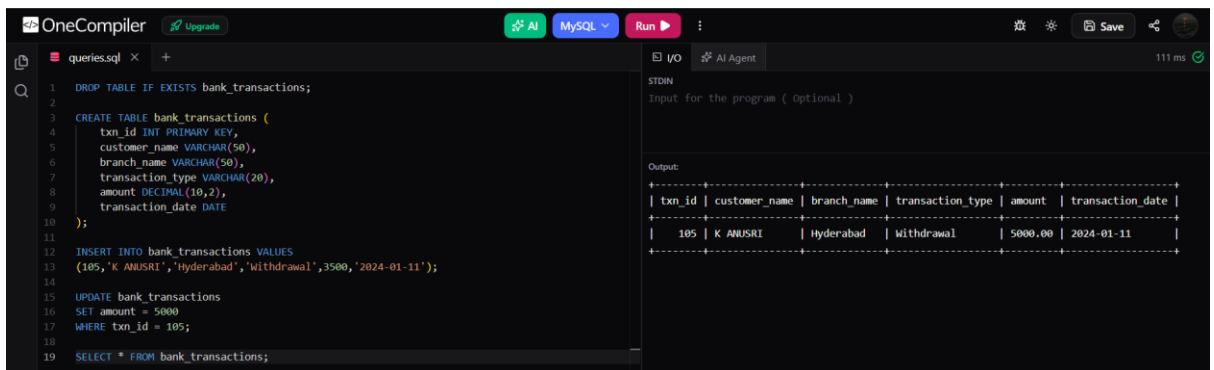
The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4   txn_id INT PRIMARY KEY,
5   customer_name VARCHAR(50),
6   branch_name VARCHAR(50),
7   transaction_type VARCHAR(20),
8   amount DECIMAL(10,2),
9   transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (111, 'K ANUSRI', 'Vizag', 'Deposit', 7000, '2024-01-18');
14
15 SELECT * FROM bank_transactions;
```

The output on the right shows a single transaction record:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
111	K ANUSRI	Vizag	Deposit	7000.00	2024-01-18

2. Update Customer Transaction Amount



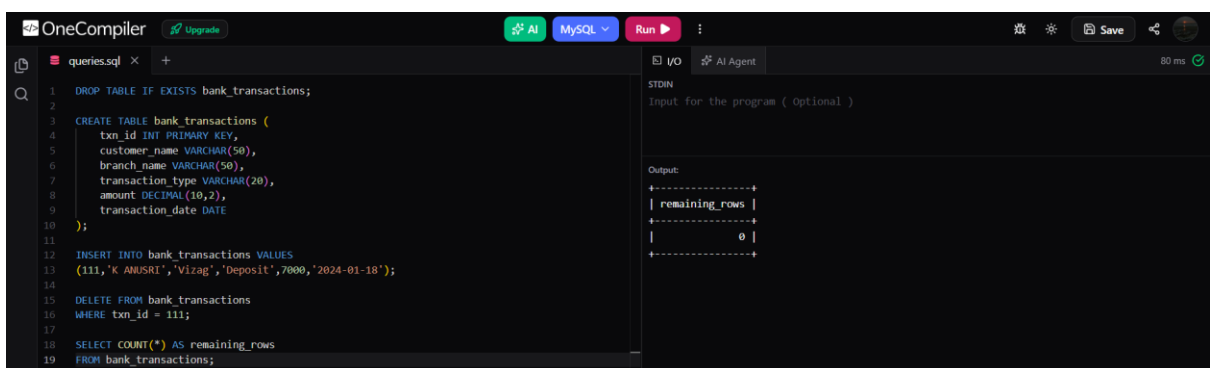
The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4   txn_id INT PRIMARY KEY,
5   customer_name VARCHAR(50),
6   branch_name VARCHAR(50),
7   transaction_type VARCHAR(20),
8   amount DECIMAL(10,2),
9   transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (105, 'K ANUSRI', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11');
14
15 UPDATE bank_transactions
16 SET amount = 5000
17 WHERE txn_id = 105;
18
19 SELECT * FROM bank_transactions;
```

The output on the right shows the updated transaction record:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
105	K ANUSRI	Hyderabad	Withdrawal	5000.00	2024-01-11

3. Delete Customer Transaction



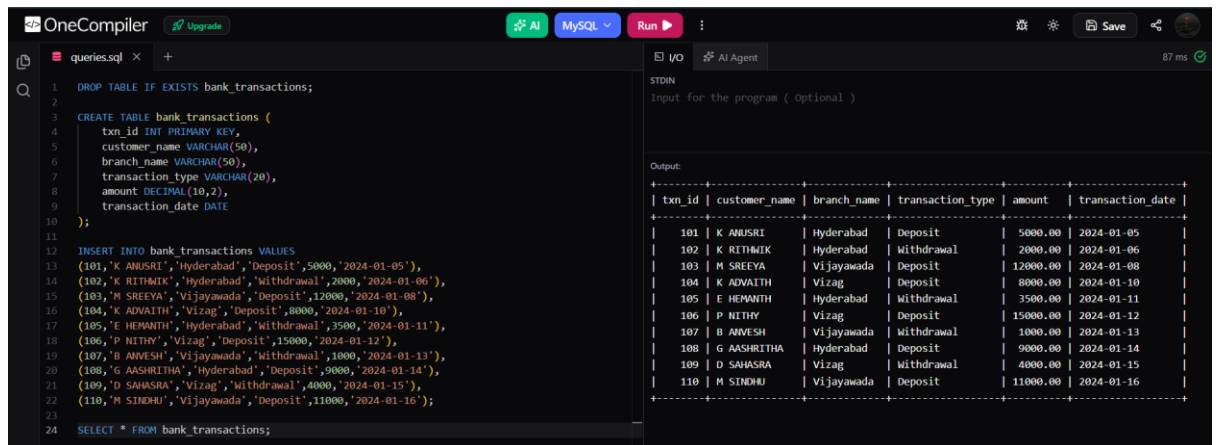
The screenshot shows the OneCompiler IDE with a MySQL database. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4   txn_id INT PRIMARY KEY,
5   customer_name VARCHAR(50),
6   branch_name VARCHAR(50),
7   transaction_type VARCHAR(20),
8   amount DECIMAL(10,2),
9   transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (111, 'K ANUSRI', 'Vizag', 'Deposit', 7000, '2024-01-18');
14
15 DELETE FROM bank_transactions
16 WHERE txn_id = 111;
17
18 SELECT COUNT(*) AS remaining_rows
19 FROM bank_transactions;
```

The output on the right shows the count of remaining rows:

remaining_rows
0

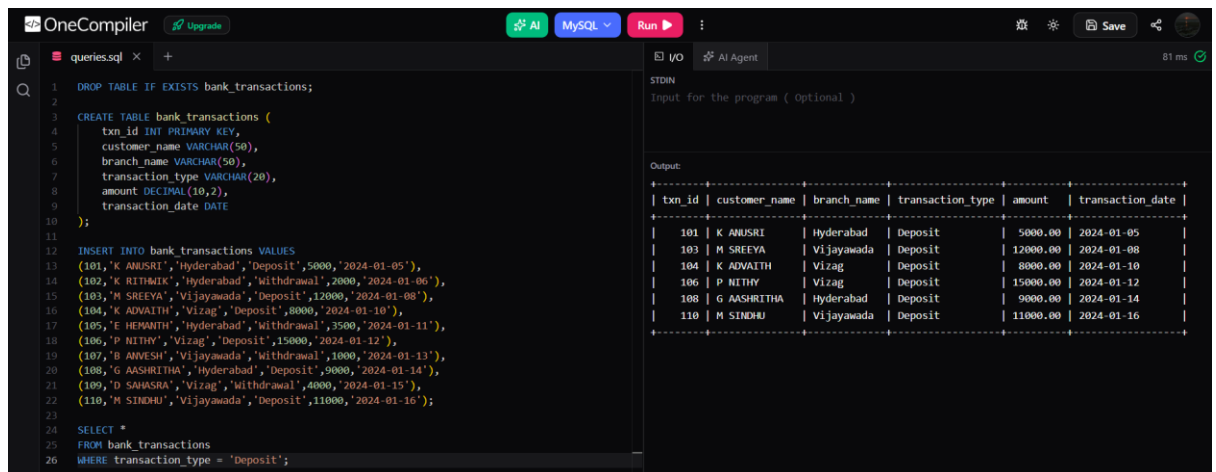
4. Display All Bank Transactions



```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (101,'K ANUSRI','Hyderabad','Deposit',5000,'2024-01-05'),
14 (102,'K RITHMIK','Hyderabad','Withdrawal',2000,'2024-01-06'),
15 (103,'M SREEYA','Vijayawada','Deposit',12000,'2024-01-08'),
16 (104,'K ADVAITH','Vizag','Deposit',8000,'2024-01-10'),
17 (105,'E HEMANTH','Hyderabad','Withdrawal',3500,'2024-01-11'),
18 (106,'P NITHY','Vizag','Deposit',15000,'2024-01-12'),
19 (107,'B ANVESH','Vijayawada','Withdrawal',1000,'2024-01-13'),
20 (108,'G AASHRITHA','Hyderabad','Deposit',9000,'2024-01-14'),
21 (109,'D SAHASRA','Vizag','Withdrawal',4000,'2024-01-15'),
22 (110,'M SINDHU','Vijayawada','Deposit',11000,'2024-01-16');
23
24 SELECT * FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	K ANUSRI	Hyderabad	Deposit	5000.00	2024-01-05
102	K RITHMIK	Hyderabad	Withdrawal	2000.00	2024-01-06
103	M SREEYA	Vijayawada	Deposit	12000.00	2024-01-08
104	K ADVAITH	Vizag	Deposit	8000.00	2024-01-10
105	E HEMANTH	Hyderabad	Withdrawal	3500.00	2024-01-11
106	P NITHY	Vizag	Deposit	15000.00	2024-01-12
107	B ANVESH	Vijayawada	Withdrawal	1000.00	2024-01-13
108	G AASHRITHA	Hyderabad	Deposit	9000.00	2024-01-14
109	D SAHASRA	Vizag	Withdrawal	4000.00	2024-01-15
110	M SINDHU	Vijayawada	Deposit	11000.00	2024-01-16

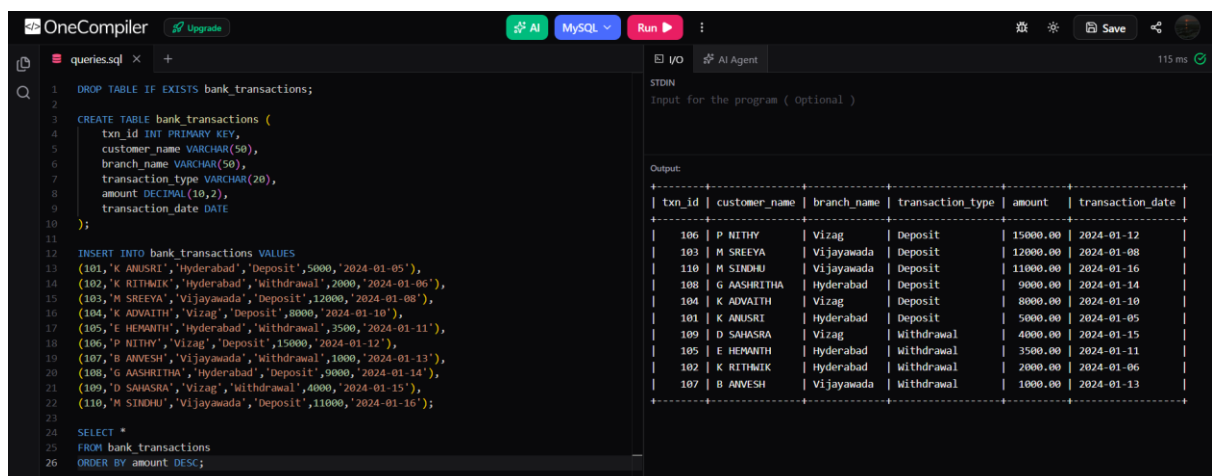
5. SELECT WITH WHERE



```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (101,'K ANUSRI','Hyderabad','Deposit',5000,'2024-01-05'),
14 (102,'K RITHMIK','Hyderabad','Withdrawal',2000,'2024-01-06'),
15 (103,'M SREEYA','Vijayawada','Deposit',12000,'2024-01-08'),
16 (104,'K ADVAITH','Vizag','Deposit',8000,'2024-01-10'),
17 (105,'E HEMANTH','Hyderabad','Withdrawal',3500,'2024-01-11'),
18 (106,'P NITHY','Vizag','Deposit',15000,'2024-01-12'),
19 (107,'B ANVESH','Vijayawada','Withdrawal',1000,'2024-01-13'),
20 (108,'G AASHRITHA','Hyderabad','Deposit',9000,'2024-01-14'),
21 (109,'D SAHASRA','Vizag','Withdrawal',4000,'2024-01-15'),
22 (110,'M SINDHU','Vijayawada','Deposit',11000,'2024-01-16');
23
24 SELECT *
25 FROM bank_transactions
26 WHERE transaction_type = 'Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	K ANUSRI	Hyderabad	Deposit	5000.00	2024-01-05
103	M SREEYA	Vijayawada	Deposit	12000.00	2024-01-08
104	K ADVAITH	Vizag	Deposit	8000.00	2024-01-10
106	P NITHY	Vizag	Deposit	15000.00	2024-01-12
108	G AASHRITHA	Hyderabad	Deposit	9000.00	2024-01-14
110	M SINDHU	Vijayawada	Deposit	11000.00	2024-01-16

6. SELECT WITH ORDER BY

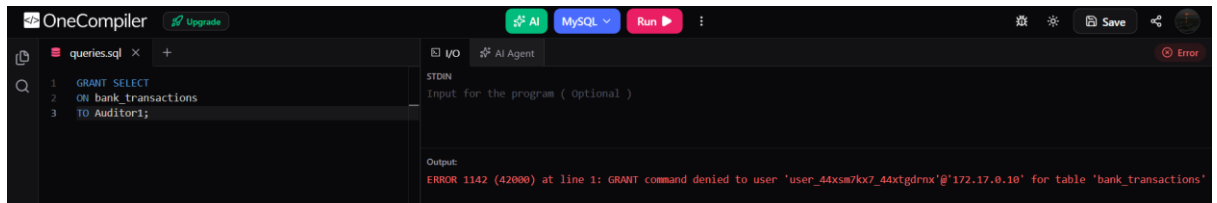


```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
12 INSERT INTO bank_transactions VALUES
13 (101,'K ANUSRI','Hyderabad','Deposit',5000,'2024-01-05'),
14 (102,'K RITHMIK','Hyderabad','Withdrawal',2000,'2024-01-06'),
15 (103,'M SREEYA','Vijayawada','Deposit',12000,'2024-01-08'),
16 (104,'K ADVAITH','Vizag','Deposit',8000,'2024-01-10'),
17 (105,'E HEMANTH','Hyderabad','Withdrawal',3500,'2024-01-11'),
18 (106,'P NITHY','Vizag','Deposit',15000,'2024-01-12'),
19 (107,'B ANVESH','Vijayawada','Withdrawal',1000,'2024-01-13'),
20 (108,'G AASHRITHA','Hyderabad','Deposit',9000,'2024-01-14'),
21 (109,'D SAHASRA','Vizag','Withdrawal',4000,'2024-01-15'),
22 (110,'M SINDHU','Vijayawada','Deposit',11000,'2024-01-16');
23
24 SELECT *
25 FROM bank_transactions
26 ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	P NITHY	Vizag	Deposit	15000.00	2024-01-12
103	M SREEYA	Vijayawada	Deposit	12000.00	2024-01-08
110	M SINDHU	Vijayawada	Deposit	11000.00	2024-01-16
108	G AASHRITHA	Hyderabad	Deposit	9000.00	2024-01-14
104	K ADVAITH	Vizag	Deposit	8000.00	2024-01-10
101	K ANUSRI	Hyderabad	Deposit	5000.00	2024-01-05
109	D SAHASRA	Vizag	Withdrawal	4000.00	2024-01-15
105	E HEMANTH	Hyderabad	Withdrawal	3500.00	2024-01-11
102	K RITHMIK	Hyderabad	Withdrawal	2000.00	2024-01-06
107	B ANVESH	Vijayawada	Withdrawal	1000.00	2024-01-13

SEC 3 — DCL COMMANDS

1. Grant SELECT Privilege to Auditor1



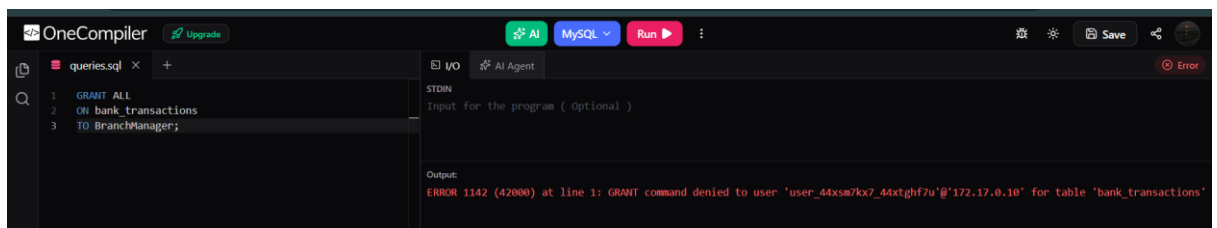
The screenshot shows the OneCompiler interface with a MySQL query in the editor:

```
1 GRANT SELECT
2 ON bank_transactions
3 TO Auditor1;
```

The output pane displays an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm7kx7_44xtgdrnx'@'172.17.0.10' for table 'bank_transactions'
```

2. Grant All Privileges to BranchManager



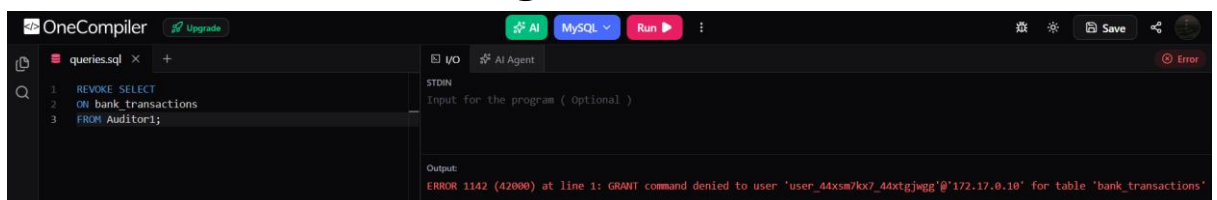
The screenshot shows the OneCompiler interface with a MySQL query in the editor:

```
1 GRANT ALL
2 ON bank_transactions
3 TO BranchManager;
```

The output pane displays an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm7kx7_44xtghf7u'@'172.17.0.10' for table 'bank_transactions'
```

3. Revoke SELECT Privilege from Auditor1



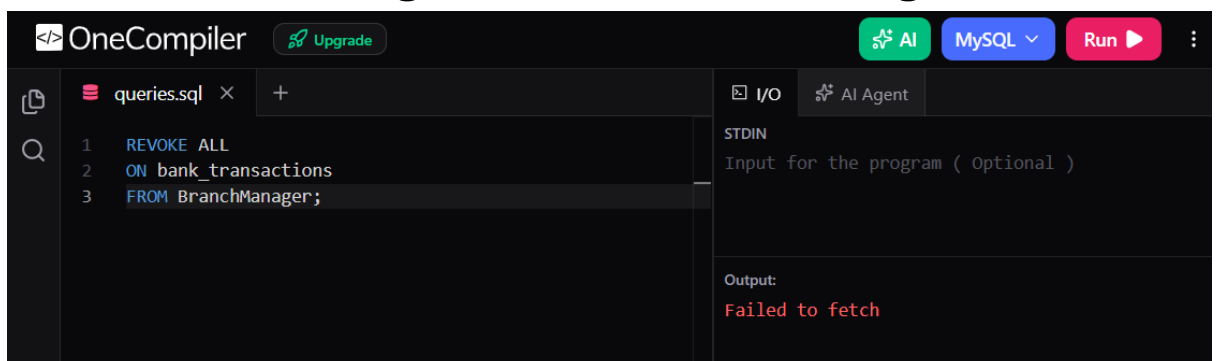
The screenshot shows the OneCompiler interface with a MySQL query in the editor:

```
1 REVOKE SELECT
2 ON bank_transactions
3 FROM Auditor1;
```

The output pane displays an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm7kx7_44xtgjjwg'@'172.17.0.10' for table 'bank_transactions'
```

4. Revoke All Privileges from BranchManager



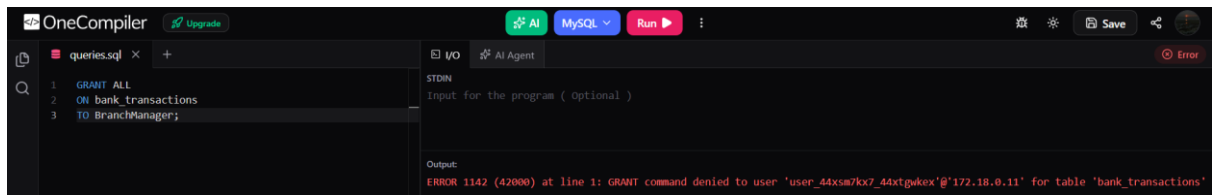
The screenshot shows the OneCompiler interface with a MySQL query in the editor:

```
1 REVOKE ALL
2 ON bank_transactions
3 FROM BranchManager;
```

The output pane displays an error message:

```
Output:
Failed to fetch
```

5. Grant All Privileges to BranchManager



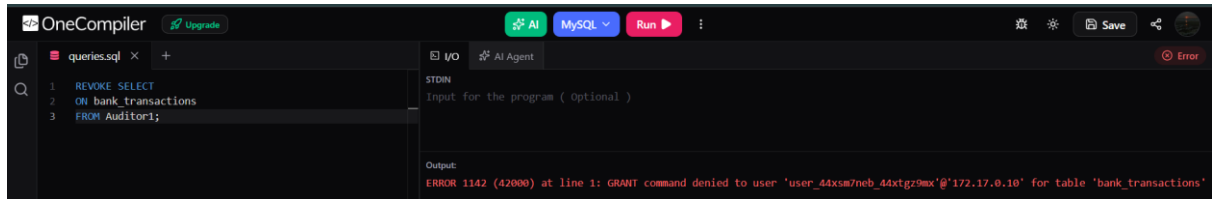
The screenshot shows the OneCompiler interface with a MySQL query in the editor. The query is:

```
1 GRANT ALL
2 ON bank_transactions
3 TO BranchManager;
```

The output pane shows an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm/kx7_44xtgwex'@'172.18.0.11' for table 'bank_transactions'
```

6. Revoke SELECT Privilege from Auditor1



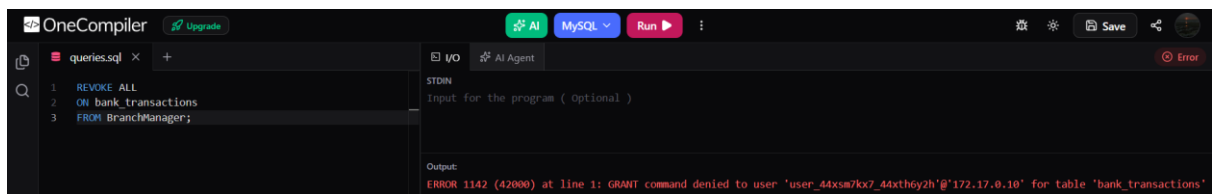
The screenshot shows the OneCompiler interface with a MySQL query in the editor. The query is:

```
1 REVOKE SELECT
2 ON bank_transactions
3 FROM Auditor1;
```

The output pane shows an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm/neb_44xtgz9mx'@'172.17.0.10' for table 'bank_transactions'
```

7. Revoke All Privileges from BranchManager



The screenshot shows the OneCompiler interface with a MySQL query in the editor. The query is:

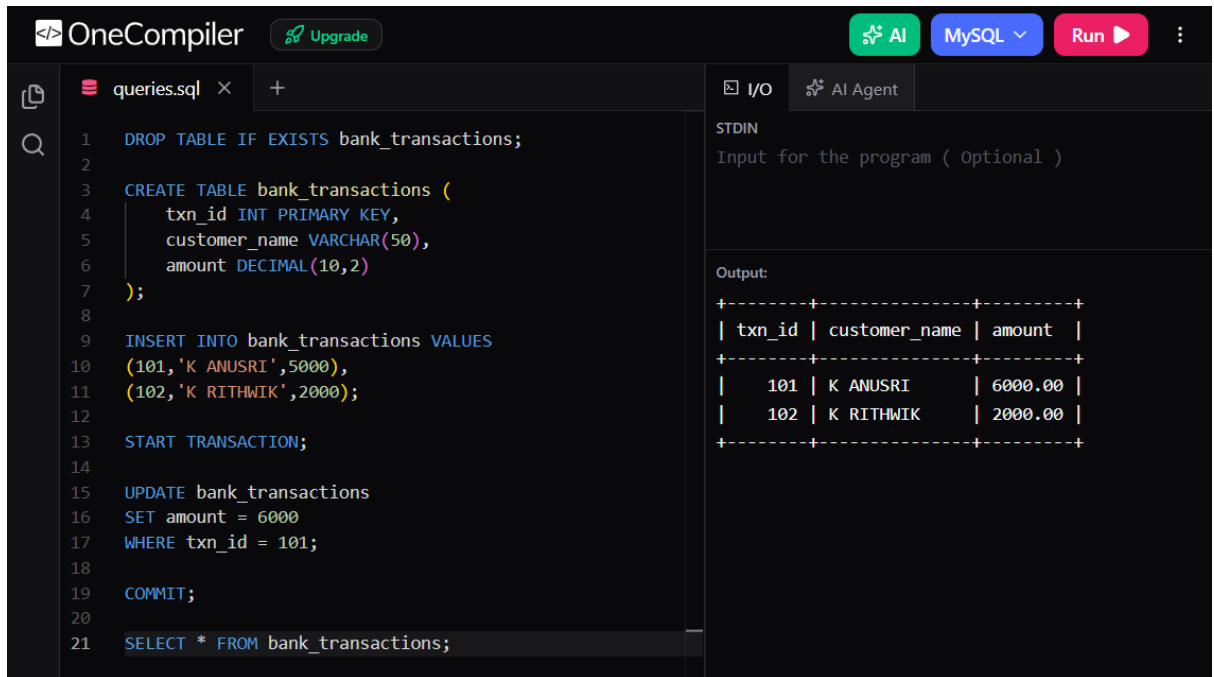
```
1 REVOKE ALL
2 ON bank_transactions
3 FROM BranchManager;
```

The output pane shows an error message:

```
Output:
ERROR 1142 (42000) at line 1: GRANT command denied to user 'user_44xsm/7kx7_44xth6y2h'@'172.17.0.10' for table 'bank_transactions'
```

SEC 4 — TCL COMMANDS

1. Commit Transaction



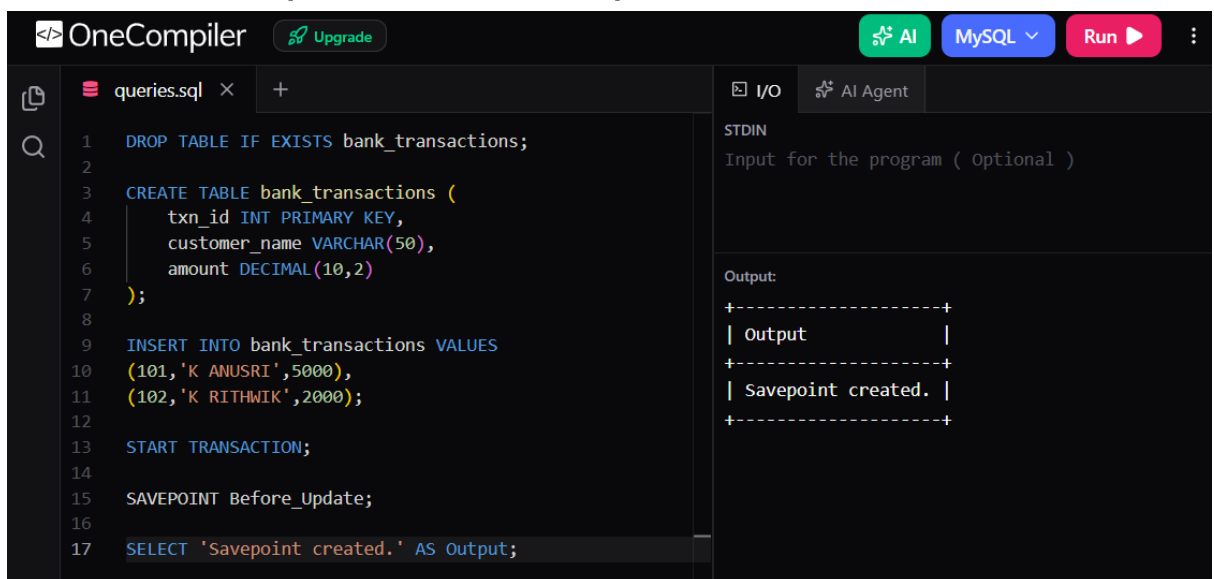
The screenshot shows the OneCompiler MySQL interface. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     amount DECIMAL(10,2)
7 );
8
9 INSERT INTO bank_transactions VALUES
10 (101, 'K ANUSRI', 5000),
11 (102, 'K RITHWIK', 2000);
12
13 START TRANSACTION;
14
15 UPDATE bank_transactions
16 SET amount = 6000
17 WHERE txn_id = 101;
18
19 COMMIT;
20
21 SELECT * FROM bank_transactions;
```

The output on the right shows the result of the query:

```
Output:
+-----+-----+-----+
| txn_id | customer_name | amount |
+-----+-----+-----+
| 101 | K ANUSRI | 6000.00 |
| 102 | K RITHWIK | 2000.00 |
+-----+-----+-----+
```

2. Create Savepoint Before Update



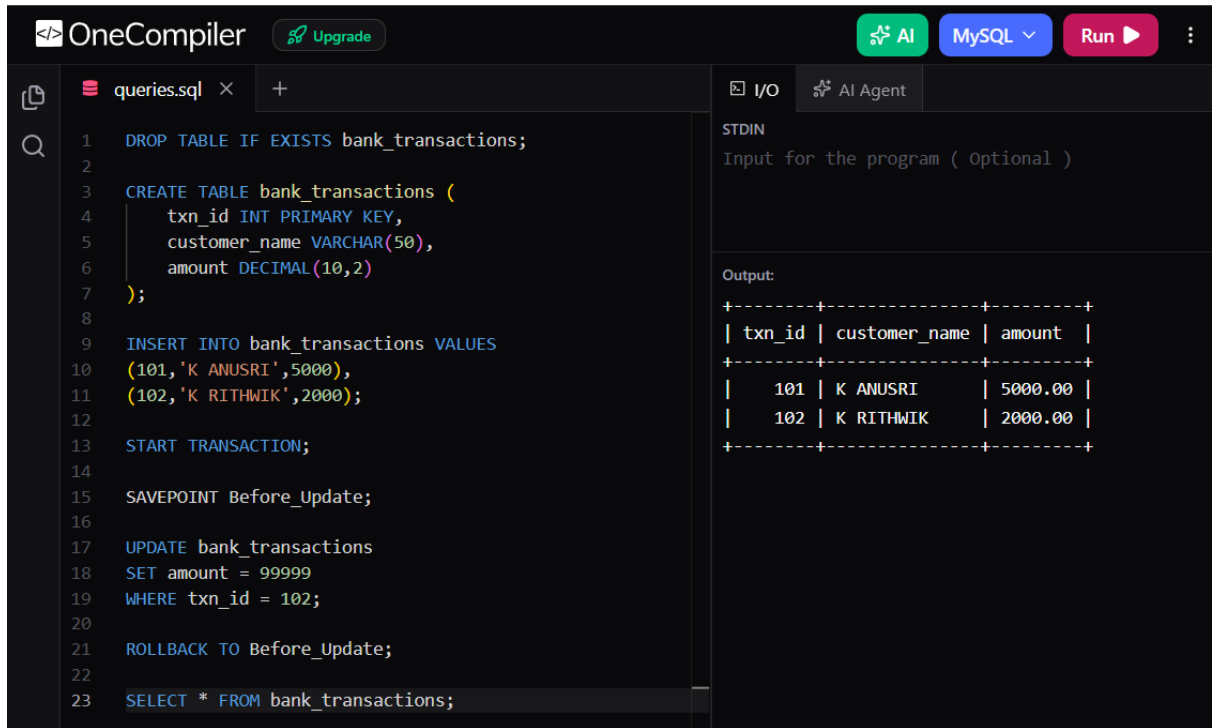
The screenshot shows the OneCompiler MySQL interface. The code in the editor is as follows:

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     amount DECIMAL(10,2)
7 );
8
9 INSERT INTO bank_transactions VALUES
10 (101, 'K ANUSRI', 5000),
11 (102, 'K RITHWIK', 2000);
12
13 START TRANSACTION;
14
15 SAVEPOINT Before_Update;
16
17 SELECT 'Savepoint created.' AS Output;
```

The output on the right shows the result of the query:

```
Output:
+-----+
| Output |
+-----+
| Savepoint created. |
+-----+
```

3. Rollback Incorrect Update to Savepoint



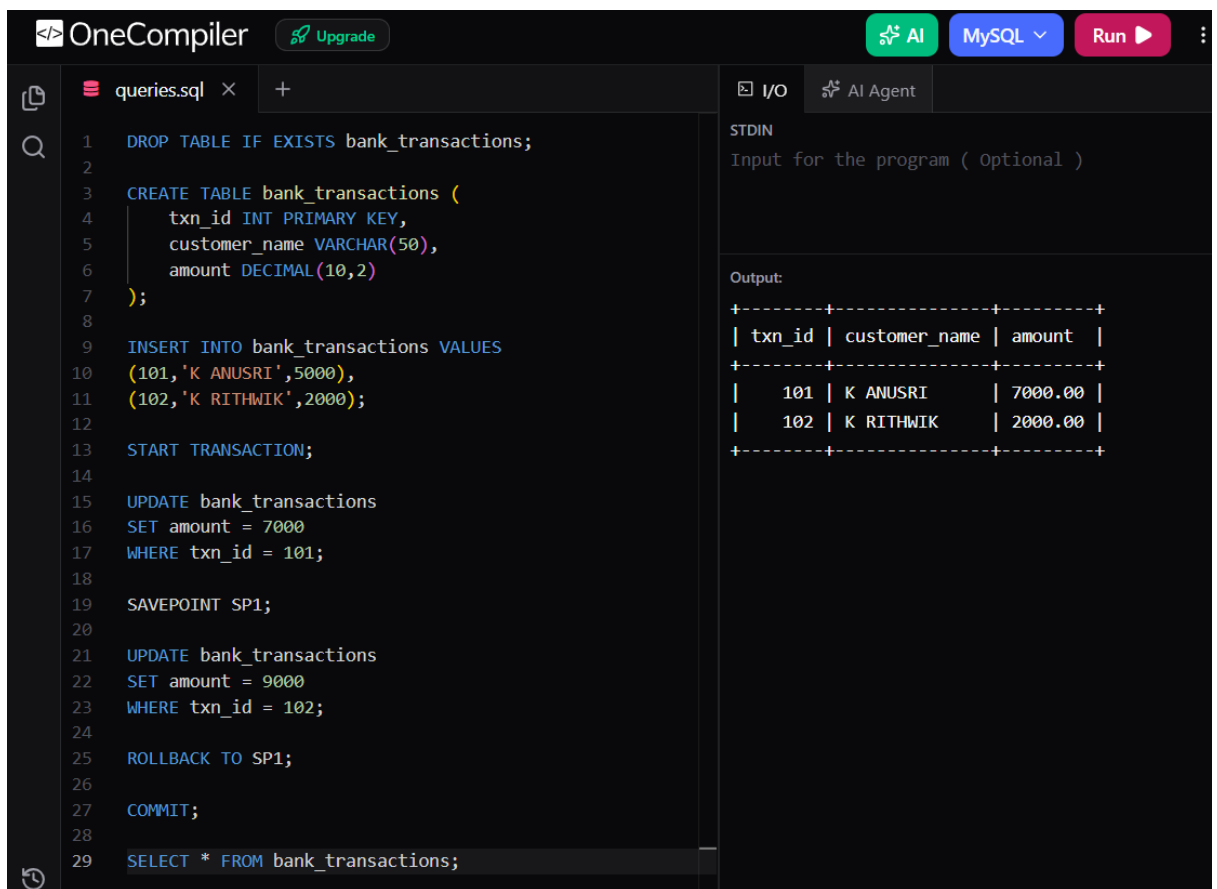
The screenshot shows the OneCompiler interface with a MySQL database selected. The SQL editor contains a script to create a table, insert data, set a savepoint, update a record, rollback to the savepoint, and select all records. The output panel shows the result of the SELECT statement as a table with two rows.

```
1 DROP TABLE IF EXISTS bank_transactions;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     amount DECIMAL(10,2)
7 );
8
9 INSERT INTO bank_transactions VALUES
10 (101, 'K ANUSRI', 5000),
11 (102, 'K RITHWIK', 2000);
12
13 START TRANSACTION;
14
15 SAVEPOINT Before_Update;
16
17 UPDATE bank_transactions
18 SET amount = 99999
19 WHERE txn_id = 102;
20
21 ROLLBACK TO Before_Update;
22
23 SELECT * FROM bank_transactions;
```

Output:

txn_id	customer_name	amount
101	K ANUSRI	5000.00
102	K RITHWIK	2000.00

4. Complete TCL Example — UPDATE, SAVEPOINT, ROLLBACK and COMMIT



The screenshot shows the OneCompiler interface with a MySQL database selected. The left pane displays a SQL script in `queries.sql` with line numbers 1 through 29. The script performs the following operations:

- 1: `DROP TABLE IF EXISTS bank_transactions;`
- 2: (blank line)
- 3: `CREATE TABLE bank_transactions (`
- 4: `txn_id INT PRIMARY KEY,`
- 5: `customer_name VARCHAR(50),`
- 6: `amount DECIMAL(10,2)`
- 7: `);`
- 8: (blank line)
- 9: `INSERT INTO bank_transactions VALUES`
- 10: `(101, 'K ANUSRI', 5000),`
- 11: `(102, 'K RITHWIK', 2000);`
- 12: (blank line)
- 13: `START TRANSACTION;`
- 14: (blank line)
- 15: `UPDATE bank_transactions`
- 16: `SET amount = 7000`
- 17: `WHERE txn_id = 101;`
- 18: (blank line)
- 19: `SAVEPOINT SP1;`
- 20: (blank line)
- 21: `UPDATE bank_transactions`
- 22: `SET amount = 9000`
- 23: `WHERE txn_id = 102;`
- 24: (blank line)
- 25: `ROLLBACK TO SP1;`
- 26: (blank line)
- 27: `COMMIT;`
- 28: (blank line)
- 29: `SELECT * FROM bank_transactions;`

The right pane shows the output of the script. It includes a section for STDIN (empty) and an "Output:" section displaying a table of the final state of the `bank_transactions` table:

txn_id	customer_name	amount
101	K ANUSRI	7000.00
102	K RITHWIK	2000.00